ChatGPT

ReefOS Species Classification & Data Model Report

Executive Summary: We propose a unified taxonomy and data schema for corals, marine fish, and reef invertebrates that serves both hobbyists and retailers. Our model combines **scientific classification** (e.g. WoRMS, FishBase) with **hobbyist groupings** (SPS/LPS/soft, reef-safe, aggression). Each species entry will include key husbandry fields (lighting, flow, diet, tank size, aggression, etc.), trade names, and care guidelines. We recommend an initial "seed set" of ~100–150 common corals (e.g. *Acropora*, *Euphyllia*, *Montipora*), ~100–150 fish (e.g. clownfish, tangs, gobies), and ~50–100 inverts (e.g. cleaner shrimp, snails, crabs) chosen by popularity and ease of care. Images must be standardized (white balance, metadata) and our workflow uses AI-assisted identification with user confirmation to grow the database. We map species to relevant products (e.g. SPS corals → high-output lights, kalk dosing; schooling fish → open water flow, filter). The database schema (see ER diagram below) includes species, classification, user inventory, and media tables. API endpoints support CRUD operations and an AI-ID flow (upload photo → get candidate species → confirm). UX flows for photo ID, onboarding new livestock, and placement assistance are specified. We also outline a moderation process for user-added entries (flagging, review, canonicalization). Finally, a phased roadmap (MVP vs v1/v2) prioritizes core fields and gradual feature rollout. Our design is based on authoritative sources like WoRMS [1] (coral taxonomy) and FishBase [2] (fish taxonomy) as well as trade experts [3] [4].

1. Taxonomy & Classification

We adopt **scientific taxonomies** from databases like WoRMS and FishBase, combined with **hobby/trade categories**. For example:

- **Corals:** Hard (Scleractinia, "stony" corals) vs Soft (Octocorallia, e.g. gorgonians) [3]. Hard corals split into **SPS (Small Polyp Stony)** and **LPS (Large Polyp Stony)** groups based on polyp size [3] [4]. WoRMS lists the full hierarchy: *Animalia* > *Cnidaria* > *Anthozoa* > *Hexacorallia* > *Scleractinia (Order)* [1]. We will store Family, Genus, Species for each coral (e.g. *Acroporidae* > *Acropora*). Trade names (e.g. "Hammer Coral" for *Euphyllia ancora*) link to species.
- **Fish:** Marine species drawn from Pomacentridae, Chaetodontidae, Tetraodontidae, etc. FishBase taxonomy (e.g. *Chordata* > *Teleostei* > *Ovalentaria* > *Pomacentridae* for clownfish) [2] will populate scientific classification. We will also tag reef-safe vs predator, schooling vs solitary, reef zone (shallow/deep) etc.
- **Invertebrates:** Key orders include Actiniaria (anemones), Decapoda (shrimps, crabs), Gastropoda (snails), Echinodermata (urchins, stars). We classify down to family/genus where relevant. Habit descriptors (e.g. "cleaner shrimp", "algae grazer") align with common names.

In addition, we incorporate hobbyist groupings (e.g. "reef-safe", "aggressive", "hardy", etc.) from sources like BulkReefSupply [3] [5]. These categories (SPS/LPS/Soft, aggressive behaviors, difficulty) will be explicit fields to aid filtering and placement decisions.

Mermaid Diagram: *Taxonomic hierarchy (sample).*

```
graph TD
  Animalia --> Cnidaria --> Anthozoa --> Hexacorallia -->
Scleractinia --> Acroporidae --> Acropora
  Animalia --> Chordata --> Teleostei --> Perciformes -->
Pomacentridae --> Pomacentrinae --> Amphiprion ocellaris
```

Figure: A reef tank with diverse SPS corals (branching Acropora*), LPS corals (leathers/polyp corals), and reef fish (surgeonfish, tangs, clownfish). Scientific classification (WoRMS) underlies all entries* [1] *.*

2. Husbandry Data Fields

Each species entry will include **structured husbandry fields** (with types/units/validation) covering care needs and constraints. Example fields by group:

- **Corals:** `lighting` (lux or PAR, e.g. 200–400 µmol), `flow` (low/med/high), `placement` (top/mid/bottom), `temperature_range` (°C), `pH_range`, `hardness_range`, `growth_rate` (slow/fast), `aggressiveness` (sweeper-tentacle reach), `difficulty` (novice/advanced) [5], `reef_zone` (shallow/deep). Validation: numeric ranges (pH 7.5–8.5, temp 24–27°C), enums (flow: Low/Mod/High), text with predefined options.
- **Fish:** `max_size` (cm), `min_tank_volume` (liters), `salinity_range` (ppt or SG), `temp_range` (°C), `diet` (carnivore/omnivore/herbivore), `activity_zone` (bottom/mid/top), `behavior` (peaceful/semiaggressive/aggressive), `reef_safe` (yes/no). E.g. Clownfish: max_size 11 cm, reef_safe: yes, aggressive: no. Validation: positive floats, enums.
- **Inverts:** `type` (shrimp/crab/snail/etc.), `use` (cleaner/grazer/detritivore), `habitat` (sand/substrate/rock), `max_size`, `salinity_range`, etc. Snails: likely herbivore, attach to rock; shrimps: carnivore/cleaner.

- **Common fields:** `care_level` (Easy/Moderate/Hard), `longevity` (years), `min_group_size` if schooling, `compatible_species_notes`.

These fields are informed by pet care guides and expert articles. For instance, SPS corals "require high-output lighting and strong flow" [5], whereas soft corals "do not require substantial flow or lighting" [6]. We will capture these as structured fields (e.g. `lighting_need = High, flow_need = High` for SPS corals).

3. Seed Species Selection

We propose an initial **seed list** drawn from trade data (popular and widely available species):

- **Corals (100–150):** Common generas like *Acropora* (SPS), *Montipora* (SPS), *Pocillopora* (SPS), *Favia/Favites* (LPS), *Euphyllia* (hammer/frogspawn, LPS), *Discosoma/Actinodiscus* (mushrooms, soft), *Sinularia/Nephthea* (soft leathers), *Xenia* (zoanthids soft), *Zoanthus* (soft zoas), *Sarcophyton* (leather), etc. Selection criteria: widespread trade availability, representation of all care levels.
- **Fish (100–150):** Common reef fish: Clownfish (*Amphiprion ocellaris/percula*), Tangs/Surgeonfish (*Paracanthurus*, *Zebrasoma*), Cardinalfish (*Apogon*), Gobies (*Gobiodon*, *Valenciennea*), Wrasses (Melanurus, Six-line), Dottybacks, Firefish, Hawksbill Rabbitfish, etc. Include popular schooling fish (Chromis) and larger species (Angelfish, Butterflyfish) with caveats. Use FishBase and trade lists to ensure coverage of each major family.
- **Inverts (50–100):** Cleanup crew & reef associates: Cleaner shrimp (*Lysmata amboinensis*), Peppermint shrimp, Hermit crabs, Astrea/Bumblebee snails, Turbo snails, Nassarius, Cerith, Feather duster worms, Sea urchins (Tuxedo urchin), Brittle stars, Bubble-tip anemone (often kept with clownfish), bubble corals (if considered invert), etc. Focus on key categories (algae eaters, detritivores, hitchhikers).

Sample lists: BulkReefSupply and other retailers frequently list their "Top 10 reef fish" or "must-have cleanup crew" (e.g. clownfish, tangs, cleaner shrimp, emerald crab) [7] [5]. We will compile definitive lists and include both scientific and common names.

4. Image Handling Guidelines

For user-uploaded photos of livestock, we will implement standardized **photo guidelines and workflows**:

- **Capture guidelines:** Advise users to use appropriate white balance (~10,000K for actinic-blue lighting), steady focus, and include a ruler or known object for scale. Prefer JPEG/PNG with embedded timestamp and geo (if available). Avoid flash reflections on glass (see reef photography best practices【42†】).
- **Metadata:** On upload, extract EXIF data (time, device) if present. Store original and generate resized versions.
- **AI Matching Workflow:** When a user uploads an image:
- **AI Analysis:** Run a trained model to suggest possible species (or genera) with confidence scores.
- **User Confirmation:** Show top 3–5 candidates with example images. E.g.:
    - *Amphiprion ocellaris (Clownfish)* – 85%
    - *Amphiprion percula (Percula Clownfish)* – 10%

- *Acanthurus leucosternon (Powder Blue Tang)* – 5%
  The user selects the correct species or "None of these."

- **Fallback:** If not found, allow user to manually enter species name (creating a pending entry).

- **Quality control:** Flag low-confidence or new entries for moderator review. Encourage multiple photos per species from different angles.

This workflow ensures high-quality data growth. For example, a clownfish photo might show multiple Amphiprion options; user selection ensures accuracy.

Figure: Clownfish (Amphiprion ocellaris) in an anemone. Our photo-ID system would show similar species and let the user confirm the match (or add a new species if none match).

5. Taxonomy-to-Product Mapping

We will link species/habitats to recommended products and services:

- **Corals → Equipment:** SPS corals map to high-output LED lights (e.g. 200–400W equivalent) and strong flow pumps, plus dosing equipment (auto-dosers for Ca/alk). LPS corals map to medium lighting and flow. Soft corals map to low-medium lighting and gentle flow. E.g. *Euphyllia* (LPS) → "T5/LED light (200 PAR), 2x powerheads 500 GPH." Soft corals (e.g. *Sinularia*) → "LED light 150 PAR, single wavemaker." [6] [5]
- **Fish → Care Supplies:** Large/active fish (tangs, large angelfish) map to large aquariums (>200L) and robust protein skimmers. Saltwater fish require marine salt mix and high-end filtration. For example, surgeonfish → "300+L tank, strong filtration, frequent water changes." Pests (if any) map to medications.

- **Inverts → Consumables:** Grazers (snails, urchins) map to algae wafers and calcareous supplements. Corallimorphs/anemones map to specialized foods. Corals-incompatible inverts (e.g. mantis shrimp) would be flagged.
- **Services:** Identify when customers might need services: e.g. *big fish purchase* → reefer pickup or local fish store, *leak risk* → recommend reef-safe leak detectors or maintenance service.

This mapping is driven by species care profiles. For instance, reef catalogs cite specific water parameters and equipment: *Acropora* (SPS) "requires 250–350 μmol·m^{-2}·s^{-1} light, Ca 400–450 ppm" [6], so product suggestions follow.

6. Database Schema & API

We propose a relational schema (Mermaid ER diagram below) including:

- **Species Table (Species):** id, scientific_name, common_name, category (Coral/Fish/Invert), family_id, genus, care_level, etc.
- **Classification Tables:** Family, Genus, with foreign keys linking Species. We may use separate tables for taxonomic consistency.
- **Livestock Table:** linking **User → Tank → Species** (with species_id FK), plus fields like given_name, acquisition_date, notes, photo_id.
- **User Table:** id, email, preferences (units, locale).
- **Image Table:** id, species_id, user_id, filepath, timestamp, meta (resolved identifications).
- **Consumables Table:** maps species_id → recommended products (foreign key to Products table).
- **Transactions/Carts:** to support buy lists.

Key relationships: - Species 1– *UserInventory* - *Family 1*– Species - User 1– *Tank 1*– Inventory

```
erDiagram
    Family ||--o{ Genus : contains
    Genus ||--o{ Species : contains
    Species ||--o{ Inventory : "owned_by"
    User ||--o{ Tank : owns
    Tank ||--o{ Inventory : "has"
    Inventory }o--|| Species : contains
    Image }o--|| Species : depicts
    Family {
      int id PK
      string name
    }
    Genus {
      int id PK
      string name
      int family_id FK
    }
    Species {
      int id PK
      string common_name
```

```
    string scientific_name
    string category  (enum: coral, fish, invert)
    int genus_id FK
    // husbandry fields...
  }
  User {
    int id PK
    string email
    // preferences...
  }
  Tank {
    int id PK
    int user_id FK
    string name
    float volume_liters
    // water params...
  }
  Inventory {
    int id PK
    int tank_id FK
    int species_id FK
    string given_name
    date acquired
  }
  Image {
    int id PK
    int species_id FK
    int user_id FK
    string url
    string meta
  }
  Product {
    int id PK
    string name
    string category
  }
  ConsumableMap {
    int species_id FK
    int product_id FK
    string relation (e.g. "needed_for")
  }
```

API endpoints (examples):
- `GET /api/species` – list or search species.
- `GET /api/species/{id}` – get details/husbandry fields.
- `POST /api/inventory` – add a species to user's tank.
- `POST /api/photos/upload` – handle image upload, trigger AI analysis.

- `POST /api/photos/confirm` – confirm species for an image.
- Standard CRUD on species (for admin) and user livestock.

7. User Flows

Photo ID Flow: User taps "Add Livestock → Upload Photo." The image is sent to an AI service which returns candidate species. We show a modal with photos/names (e.g. multiple clownfish variants). The user selects one or "Not listed." If unlisted, a form appears to enter a name and details. Confirmed entries link to the inventory.

Onboarding Species (Wizard): On creating a new tank, prompt for environment (reef/mixed/fish-only), lighting type, water volume, target livestock type (soft/LPS/SPS). This pre-populates default parameters.

Placement Assistant: When adding a new coral, use its `flow` and `aggression` fields to highlight "zones" in the tank: e.g. suggest top-front for SPS, away from fast-growers, with live alerts like "Maintain 5″ spacing from nearest Acropora."

Custom Species: If a user has a unique specimen, they tap "Add New Species" and fill in basic data (common name, category). This creates a pending Species entry marked for review.

These flows are illustrated below:

```
graph LR
  A[Upload Tank Photo] --> B(Analyze with AI)
  B --> C{Confident Match?}
  C -- Yes --> D[Show candidate species list]
  C -- No --> E[User enters species manually]
  D --> F[User selects species]
  F --> G[Assign to inventory]
  E --> G
```

8. Moderation & Curation

User-added entries (new species or photos) enter a **pending queue**. Moderators (or automated checks) must verify:
- Check that species names match recognized taxonomy (using WoRMS/FishBase APIs if needed).
- Review photos: ensure correct labeling.
- Merge duplicates (e.g. user added "Percula Clown" vs *Amphiprion percula*).
Approved entries are promoted to the canonical database.

We recommend a lightweight role system: users earn trust and can help flag errors. All changes are logged.

9. Roadmap & Effort

MVP: Core tables (User, Tank, Species, Inventory), essential fields, photo upload & AI-match, basic taxonomies (~50 species).
v1: Full seed dataset, advanced husbandry fields, placement logic, product mapping, community entries.
v2: Marketplace integration, notifications, advanced AI (disease detection), multilingual support.

Effort estimates:
- DB and API (MVP): 2–3 weeks.
- Frontend flows (MVP): 2 weeks.
- Data entry (seed species, taxonomy): 4+ weeks (can be parallelized).
- AI integration and testing: 3 weeks.

10. Sources (taxonomies & care)

- **WoRMS** – World Register of Marine Species (coral and invertebrate taxonomy) [1].
- **FishBase** – Fish species data and classification [2].
- **BulkReefSupply** – reef hobby care guides (Soft vs SPS/LPS coral care) [3] [5].
- **FishBase Aquarium Summaries** – e.g. Amphiprion ocellaris profile for trade status [7].
- **ITIS / CoL** – taxonomic validation (linked via FishBase).
- **Trade publications** (Tropical Fish Hobbyist, Reef2Reef) for husbandry trends and product mapping.
- **NSF/peer-reviewed** coral catalogs (scleractinian species lists).
- **APPA Pet Stats** for market data (background).

All decisions are grounded in these authoritative classifications and practical care knowledge [1] [3] [5].

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[1] WoRMS - World Register of Marine Species - Scleractinia
https://www.marinespecies.org/aphia.php?p=taxdetails&id=1363

[2] [7] Amphiprion ocellaris, Clown anemonefish : aquarium
https://www.fishbase.se/summary/amphiprion-ocellaris.html

[3] [4] [5] [6] What Is the Difference Between Stony & Soft Corals? - Bulk Reef Supply
https://www.bulkreefsupply.com/content/post/what-is-the-difference-between-soft-and-stony-corals?srsltid=AfmBOoqwhdG39zsOuH6ZsITyFN0dLilV1gOETrL7gNGJQkQ8WXacYPBz